

3(a)	no <u>net</u> thermal energy is transferred (between them)	B1
3(b)(i)	variation (of density with temperature) is linear or each temperature has a unique value of density	B1
3(b)(ii)	- variation (of density with temperature) is not linear - region where the density does not vary with temperature - different temperatures have the same density <i>Any two points, 1 mark each</i>	B2
3(c)(i)	boiling point = 80 °C	A1
3(c)(ii)	$Q = Pt$ and $t = 21 \text{ s}$ (thermal energy supplied = $810 \times 21 = 17000 \text{ J}$)	C1
	$c = Q / m\Delta\theta$	C1
	thermal energy absorbed by beaker = $42 \times 0.84 \times (80 - 25)$ (= 1940 J)	C1
	s.h.c. of liquid = $[(810 \times 21) - (42 \times 0.84 \times (80 - 25))] / [120 \times (80 - 25)]$ = $2.3 \text{ J g}^{-1} \text{ K}^{-1}$	A1
3(d)	sketch: straight diagonal line from 25 °C to 100 °C and then horizontal at 100 °C	B1
	straight diagonal line starting at 25 °C with gradient approximately half that of the original line	B1